

$$z = f(x, y)$$

$$p = \frac{\partial z}{\partial x}$$

$$q = \frac{\partial z}{\partial y}$$

$$r = \frac{\partial^2 z}{\partial x^2}$$

$$t = \frac{\partial^2 z}{\partial y^2}$$

$$s = \frac{\partial^2 z}{\partial x \partial y}$$

Partial differential Equation :-

A partial differential equation is a relation between independent variables, dependent variable and partial derivatives of dependent variable with respect to independent variable.

Ex :  $x^2 p + y^2 q = z$



Q Find Partial differential equation from.  
 $\phi(x+y+z, x^2+y^2-z^2)=0$

Solution:- let,  $x+y+z = u$   
 $x^2+y^2-z^2 = v$

$$\therefore \phi(u, v) = 0 \quad \text{--- (1)}$$

Differentiating partially with respect to  $x$ .

$$\frac{\partial \phi}{\partial u} \left( \frac{\partial u}{\partial x} + p \frac{\partial v}{\partial x} \right) + \frac{\partial \phi}{\partial v} \left( \frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0$$

$$\Rightarrow \frac{\partial \phi}{\partial u} (1+p) + \frac{\partial \phi}{\partial v} (2x - 2zp) = 0$$

Again differentiating partially with respect to  $y$ .

$$\frac{\partial \phi}{\partial u} (1+q) + \frac{\partial \phi}{\partial v} (2y - 2zq) = 0$$

Eliminating  $\frac{\partial \phi}{\partial v}$  and  $\frac{\partial \phi}{\partial u}$ , we get ---

$$(1+p)(2y - 2zq) - (1+q)(2x - 2zp) = 0$$

$$\Rightarrow (y+z)p - (x+z)q = x-y.$$

## Lagrange methode :

\* Lagrange Auxiliary Equation

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

$$\rightarrow Pp + Qq = R$$

$$\text{Ex: } x^2 \frac{\partial z}{\partial x} + 5x \frac{\partial z}{\partial y} = 9$$

$$\Rightarrow x^2 p + 5x q = 9$$

$$\Rightarrow Pp + Qq = R$$

$$\left[ \begin{array}{l} x^2 = P \\ 5x = Q \\ 9 = R \end{array} \right]$$

1. Solve  $xzp + yzq = xz$  by L.M.

Solution :

Given  $xzp + yzq = xz$

$$\therefore P = xz$$

$$Q = yz$$

$$R = xz$$

$\therefore$  Lagrange Auxiliary Equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$$

From 1st and 2nd.

$$\frac{dz}{x^2} = \frac{dy}{y^2}$$

$$\Rightarrow \frac{dz}{x} = \frac{dy}{y}$$

Integrating both sides, we get...

$$\int \frac{dz}{x} = \int \frac{dy}{y}$$

$$\Rightarrow \ln x = \ln y + \ln C_1$$

$$\Rightarrow \ln x - \ln y = \ln C_1$$

$$\Rightarrow \ln\left(\frac{x}{y}\right) = \ln C_1$$

$$\therefore C_1 = \frac{x}{y}$$

From 1st and 3rd.

$$\frac{dx}{x^2} = \frac{dz}{z^2}$$

$$\Rightarrow dx = dz$$

Integrating both side, we get

$$\int dx = \int dz$$

$$\Rightarrow x = z + C_2$$

$$\therefore C_2 = x - z$$

Thus general Solution is

$$\phi(C_1 + C_2) = 0$$

$$\text{i.e., } \phi\left(\frac{x}{y}, x - z\right) = 0$$

(Ans.)

Problem 2: Solve  $\frac{y^2 z}{x} p + xzq = y^2$  by L.M.

Solution, Given,  $\frac{y^2 z}{x} p + xzq = y^2$

$$\therefore P = \frac{y^2 z}{x}$$

$$Q = xz$$

$$R = y^2$$

$\therefore$  Lagrange Auxiliary Equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}$$

$$\Rightarrow \frac{x dx}{y^2 z} = \frac{dy}{x^2} = \frac{dz}{y^2}$$

From 1st two, we get...

$$\frac{x dx}{y^2 z} = \frac{dy}{x^2}$$

$$\Rightarrow \frac{x dx}{y^2} = \frac{dy}{x}$$

$$\Rightarrow x^2 dx = y^2 dy$$

$$\therefore \int x^2 dx = \int y^2 dy$$

$$\Rightarrow \frac{x^3}{3} = \frac{y^3}{3} + \frac{C_1}{3}$$

$$\therefore C = x^3 - y^3$$

[Integrating]

From 1st and third...

$$\frac{x dx}{y^2 z} = \frac{dz}{y^2}$$

$$\Rightarrow x dx = z dz$$

$$\therefore \int x dx = \int z dz$$

$$\frac{x^2}{2} = \frac{z^2}{2} + \frac{C_2}{2}$$

$$\therefore c_2 = x^2 - y^2$$

Thus general solution is

$$\phi(c_1, c_2) = 0$$

$$\text{i.e., } \phi(x^3 - y^3, x^2 - y^2) = 0$$

## Lecture 2

# Solve  $(x^2 - yz)P + (y^2 - zx)Q = z^2 - xy$  by L.M.

Solution :-

$$\text{Given, } (x^2 - yz)P + (y^2 - zx)Q = z^2 - xy$$

$\therefore$  Lagrang Auxiliary Equation are,

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$$

$$\Rightarrow \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} = \frac{dx - dy}{x^2 - yz - y^2 + xy} = \frac{dy - dz}{y^2 - zx - z^2 + zy} = \frac{dz - dx}{z^2 - xy - x^2 + yz}$$

$$\Rightarrow \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} = \frac{dx - dy}{x - y} = \frac{dy - dz}{y - z} = \frac{dz - dx}{z - x}$$

From 4th & 5th

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$$

Integrating, we get.

$$\int \frac{dx-dy}{x-y} = \int \frac{dy-dz}{y-z}$$

$$\Rightarrow \ln(x-y) = \ln(y-z) + \ln c_1$$

$$\Rightarrow \ln(x-y) - \ln(y-z) = \ln c_1$$

$$\Rightarrow \ln \frac{(x-y)}{(y-z)} = \ln c_1$$

$$\therefore c_1 = \frac{x-y}{y-z}$$

From last two ...  $\frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$

Now Integrating, we get...

$$\int \frac{dy-dz}{y-z} = \int \frac{dz-dx}{z-x}$$

$$\therefore c_2 = \frac{y-z}{z-x}$$

$\therefore$  Complete solution is  $\phi(c_1, c_2) = 0$

$$\text{i.e., } \phi\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0$$

Ans.

Equation coefficient of  $x^m$  from both side.

$$m(m-1)a_0 - ma_0 + a_0 = 0$$

$$\Rightarrow a_0 (m^2 - m - m + 1) = 0$$

$$\Rightarrow a_0 (m^2 - 2m + 1) = 0$$

$$\therefore m^2 - 2m + 1 = 0 \quad [\because a_0 \neq 0]$$

$$\Rightarrow (m-1)^2 = 0$$

$$\therefore m = 1, 1$$

Here,  $m_1 = 1$ ,  $m_2 = 1$ .

i.e. complete solution is.

$$y = C_1 (x)^{m=m_1} + C_2 (x)^{m=m_2} \quad \text{--- (iv)}$$

Equating coefficient of  $x^{m+1}$  we get.

$$(m+1)ma_1 + (m+1)a_1 + a_1 = 0$$

$$\Rightarrow a_1 (m^2 + m - m - 1 + 1) = 0$$

$$\Rightarrow a_1 m^2 = 0$$

$$\therefore a_1 = 0 \quad [\because m \neq 0]$$

Equating coefficient of  $x^{m+2}$ , we get...

$$(m+1)(m+2)a_2 - (m+2)a_2 + a_0 + a_2 = 0$$

$$\Rightarrow a_2 (m^2 + 2m + m + 2 - m - 2 + 1) = -a_0$$

$$\Rightarrow a_2 (m^2 + 2m + 1) = -a_0$$

$$\therefore a_2 = \frac{a_0}{(m+1)^2}$$

Putting  $a_1, a_2, \dots$  in equation (11)

$$y = a_0 x^m + 0 \cdot x^{m+1} - \frac{a_0}{(m+1)^2} x^{m+2} + \dots$$

$$= a_0 x^m - \frac{a_0}{(m+1)^2} x^{m+2} + \dots$$

Again,  $\frac{\partial y}{\partial x} = m a_0 x^{m-1} - \frac{(m+2) a_0}{(m+1)^2} x^{m+1} + \dots$

Now,  $(y)_{m=m_1} = a_0 x - \frac{a_0}{(m+1)^2} x^3 + \dots$

and,  $(\frac{\partial y}{\partial x})_{m=m_2} = a_0 x^0 - \frac{3}{4} x^2 + \dots$

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#  $(x^2+1)y'' + xy' + 2xy = 0$  solve by series method where  $y(0) = 1$  and  $y'(0) = 1$

Solution:-

Given DE =  $(x^2+1)y'' + xy' + 2xy = 0$  ①

Here,  $P(x) = \frac{x^2+1}{x^2+1} = 1 \neq 0$ .

i.e.,  $x=0$  is an ordinary point.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} + \dots$$

$$y'' = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots$$

Putting in ①.

$$\begin{aligned} & 2a_2x^2 + 6a_3x^3 + \dots + n(n-1)a_nx^n + \dots + 2a_2 + 6a_3x + \dots \\ & + n(n-1)a_nx^{n-2} + \dots + a_1x + 2a_2x^2 + 3a_3x^3 + \dots \\ & + na_nx^n + \dots + 2a_0x + 2a_1x^2 + 2a_2x^3 + \dots + 2a_nx^{n+1} = 0 \end{aligned}$$

Equating coefficient of  $x^0$ .

$$2a_2 = 0$$

$$\therefore a_2 = 0$$

Equating coefficient of  $x$

$$6a_3 + a_1 + 2a_0 = 0$$

$$\Rightarrow 6a_3 = -a_1 - 2a_0$$

$$\therefore a_3 = \frac{-a_1 - 2a_0}{6}$$

Equating coefficient of  $x^n$

$$n(n-1)a_n + (n+2)(n+1)a_{n+2} + na_n + 2a_{n-1} = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} = -n(n-1)a_n - na_n - 2a_{n-1}$$

$$= -n^2a_n + na_n - na_n - 2a_{n-1}$$

$$= -n^2a_n - 2a_{n-1}$$

$$\therefore a_{n+2} = \frac{-n^2a_n - 2a_{n-1}}{(n+2)(n+1)}$$

Putting  $n = 2, 3, 4, \dots$

$$a_4 = \frac{-4a_2 - 2a_1}{12}$$

$$= -\frac{2a_1}{12} = -\frac{a_1}{6}$$

$$a_5 = \frac{-9a_3 - 2a_2}{20}$$

$$= \frac{-9a_3}{20} = -\frac{9}{20} \cdot \frac{-a_1 - 2a_0}{6} = \frac{3a_1 + 6a_0}{40}$$

Impedant Solve  $(y+z)p + (z+x)q = x+y$

Solution:

Given,  $(y+z)p + (z+x)q = x+y$

Lagrange Auxiliary Equation are.

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$$

$$\Rightarrow \frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{dx-dy}{y-z} = \frac{dy-dz}{z-y} = \frac{dz+dx}{x-z}$$

From 4th & 5th.

$$\frac{dx-dy}{y-z} = \frac{dy-dz}{z-y}$$

$$\Rightarrow \frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$$

Integrating, we get.

$$\int \frac{dx-dy}{x-y} = \int \frac{dy-dz}{y-z}$$

$$\Rightarrow \ln(x-y) = \ln(y-z) + \ln C_1$$

$$\Rightarrow \ln \left( \frac{x-y}{y-z} \right) = \ln C_1$$

$$\therefore C_1 = \frac{x-y}{y-z}$$

From last too

$$\frac{dy-dz}{z-y} = \frac{dz-dx}{x-z}$$

$$\Rightarrow \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}$$

Integrating, we get.

$$\int \frac{dy-dz}{y-z} = \int \frac{dz-dx}{z-x}$$

$$\Rightarrow \ln(y-z) = \ln(z-x) + \ln C_2$$

$$\Rightarrow \ln \frac{(y-z)}{(z-x)} = \ln C_2$$

$$\therefore C_2 = \frac{y-z}{z-x}$$

Complete solution is  $\phi(C_1, C_2) = 0$

$$\text{i.e., } \phi\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0 \quad (\text{Ans...})$$

## Charpit's auxiliary Equation

$$\frac{dp}{\frac{\partial f}{\partial p} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial q} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$

Q Find the complete integral of the equation.  
 $Px + Qy = P^2$  by Charpit's method.

Solution:

Given,  $Px + Qy = P^2$

$$\Rightarrow Px + Qy - P^2 = 0$$

$$\therefore f = Px + Qy - P^2$$

Now,  $\frac{\partial f}{\partial x} = P$

$$\frac{\partial f}{\partial y} = Q$$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial p} = x - 2P$$

$$\frac{\partial f}{\partial q} = y - P$$

Charpit's auxiliary equations are,

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial p}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial q}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

$$\Rightarrow \frac{dp}{p} = \frac{dq}{q} = \frac{dz}{-px - qy + 2pq} = \frac{dx}{q-x} = \frac{dy}{p-y} = \frac{dF}{0}$$

From first two,

$$\frac{dp}{p} = \frac{dq}{q}$$

Integrating, we get.

$$\ln p = \ln q + \ln a$$

$$\therefore a = \frac{p}{q} \quad \text{--- (1)}$$

$\Rightarrow p = aq$  [where  $a$  is arbitrary constant]

putting  $p = aq$  in given equation.

$$q(ax+y) = aq^2$$

$$\Rightarrow aq = ax+y$$

$$\therefore q = \frac{ax+y}{a}$$

$$dz = p dx + q dy$$

From ①:

$$p = a q$$

$$= a \cdot \frac{ax+y}{a}$$

$$= ax+y$$

putting these value into

$$dz = p dx + q dy$$

$$\Rightarrow dz = (ax+y) dx + \left(\frac{ax+y}{a}\right) dy$$

$$\Rightarrow a dz = (y+ax)(dy+adx)$$

Integrating, we get,

$$az = \frac{1}{2} (y+ax)^2 + b$$

which is required complete solution, where  $b$  is an arbitrary constant.

(5)

# Solve  $2xz - px^2 - 2qxy + pq = 0$  by Charpit's method.

Solution:

Given:  $2xz - px^2 - 2qxy + pq = 0$  (1)

$$f(x, y, z, p, q) = 2xz - px^2 - 2qxy + pq$$

Now,  $\frac{\partial f}{\partial z} = 2x - 2px - 2qy$

$$\frac{\partial f}{\partial y} = -2qx$$

$$\frac{\partial f}{\partial z} = 2x$$

$$\frac{\partial f}{\partial p} = -x^2 + q$$

$$\frac{\partial f}{\partial q} = -2xy + p$$

From Charpit auxiliary equations, we get.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$

$$\Rightarrow \frac{dp}{2x - 2qy} = \frac{dq}{0} = \frac{dz}{px^2 - 2xyq - 2pq} = \frac{dx}{x^2 - q} = \frac{dy}{2xy + p} = \frac{df}{0}$$

From second, we get,

$$dq = 0$$

Integrating we get,

$$\int dq = 0$$

$$\Rightarrow q = c$$

where  $c$  is an arbitrary constant.

putting  $q = c$  in (1), we get.

$$2xz - px^2 - 2cxy + pc = 0$$

$$p(c - x^2) = 2cxy - 2xz$$

$$\therefore p = \frac{2x(cy - z)}{c - x^2}$$

$$= \frac{2x(z - cy)}{x^2 - c}$$

From  $dz = p dx + q dy$ , we get...

$$dz = \frac{2x(z - cy)}{x^2 - c} dx + c dy$$

$$\Rightarrow dz - c dy = \frac{2x(z - cy)}{x^2 - c} dx$$

$$\Rightarrow \frac{dz - c dy}{z - cy} = \frac{2x dx}{x^2 - c}$$

$$\Rightarrow \ln(z - cy) = \ln(x^2 - c) + \ln b$$

[where  $b$  is an arbitrary constant]  
which is the complete solution.

$$\text{let, } x^2 - c = m$$

$$2x dx = dm$$

$$\therefore \int \frac{dm}{m} = \ln m$$

$$\text{let, } z - cy = m$$

$$dz - c dy = dm$$

$$\therefore \int \frac{dm}{m} = \ln m$$

Charpit's method এর সমাধান করার নিয়ম :

- ① প্রথমে  $f$  এর কন্ডে নিতে হবে।
- ②  $f$  কে  $x, y, z, p, q$  এর সাপেক্ষে partial derivative করতে হবে।
- ③  $f$  এর derivativeগুলোকে Charpit's auxiliary equation বসাতে হবে।
- ④  $p, q$  এর মান বের করতে হবে।
- ⑤  $dz = p dx + q dy$  সমীকরণে  $p, q$  এর মান বসিয়ে Integration করতে হবে।

# Solve  $q = -xp + p^2$  by charpit's method.

Solution:-

Given  $q = -xp + p^2$

$$\Rightarrow q + xp - p^2 = 0 \quad \text{--- (1)}$$

$$f(x, y, z, p, q) = q + xp - p^2$$

Now,

$$\frac{\partial f}{\partial x} = p$$

$$\frac{\partial f}{\partial y} = 0$$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial p} = x - 2p$$

$$\frac{\partial f}{\partial q} = 1$$

From charpit's auxiliary equation, we get,

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} =$$

~~From 2nd we get,~~

$$\Rightarrow \frac{dp}{p} = \frac{dq}{0} = \frac{dz}{-px + 2p^2 - q} = \frac{dx}{2p - x} = \frac{dy}{-1}$$

From end. we get.

$$dq = 0$$

Integrating we get...

$$q = c.$$

putting  $q = c$  in (1) we get.

$$c + xp - p^2 = 0$$

$$\Rightarrow p^2 - xp - c = 0$$

$$\therefore p = \frac{x \pm \sqrt{x^2 + 4c}}{2}$$

From  $dz = p dx + q dy$  we get...

$$dz = \frac{x \pm \sqrt{x^2 + 4c}}{2} dx + c dy.$$

Integrating, we get...

$$z = \frac{1}{4} \left[ x^2 \pm x \sqrt{x^2 + 4c} + c \log(x + \sqrt{x^2 + 4c}) \right] + cy + b$$

where  $b$  is an arbitrary constant  
which is the complete solution

$$\begin{aligned} & \int \sqrt{x^2 + a^2} dx \\ &= x \sqrt{x^2 + a^2} + a \log(x + \sqrt{x^2 + a^2}). \end{aligned}$$

## Solution By Series Method :

Ordinary point :- If  $P_0(x)y'' + P_1(x)y' + P_2(x)y = 0$  is a differential equation then  $x = x_0$  is called ordinary point if  $P_0(x) \neq 0$  at  $x = x_0$ .

Example :- Consider the differential equation.

$$(x^2+1)y'' + xy' + y = 0$$

Here  $P_0(x) = x^2+1$ .

at,  $x=0$ ,  $P_0(0) = 0^2+1$   
 $= 1 \neq 0$

i.e.,  $x=0$  is a ordinary point of Given equation

Singular Point :- If  $P_0(x)y'' + P_1(x)y' + P_2(x)y = 0$  is a differential equation then  $x = x_0$  is called singular point if  $P_0(x) = 0$  at  $x = x_0$ .

Example :- Consider the differential equation.

$$x^2y'' + (x+1)y' + xy = 0$$

Here,  $P_0(x) = x$

$\therefore P_0(0) = 0$

i.e.,  $x=0$  is a singular point of given differential equation.

# Solve  $y' + xy' + y = 0$  by series method.

Sol<sup>n</sup>: Given differential equation is.

$$y' + xy' + y = 0 \quad \text{--- (1)}$$

Here,  $P_0(x) = 1$ . and  $x=0$  is an ordinary point of (1).

Consider a series solution of (1)

$$y = \sum_{n=0}^{\infty} a_n x^n \quad \text{--- (ii)}$$

Now,  $y' = \sum_{n=0}^{\infty} n a_n x^{n-1}$

Now  $y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$

putting the value of  $y, y', y''$  in equation ①

$$\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} + x \sum_{n=0}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \{ n(n-1) a_n x^{n-2} + n a_n x^n + a_n x^n \} = 0$$

Recurrence formula

$$(n+2)(n+1) a_{n+2} + n a_n + a_n = 0$$

$[n = n+2 \text{ বসিয়ে}]$

$$\Rightarrow a_{n+2} = \frac{-n a_n - a_n}{(n+2)(n+1)}$$

→ সবথেকে বড়টিকে নিতে হবে।

Putting  $n=0$ ,

$$a_2 = \frac{-0 \cdot a_0 + a_0}{2} = -\frac{a_0}{2}$$

$$n=1, \quad a_3 = \frac{-a_1 - a_1}{6} = -\frac{a_1}{3}$$

$$n=2, \quad a_4 = \frac{-2a_2}{12} = -\frac{1}{4} \cdot \left(-\frac{a_0}{2}\right) = \frac{a_0}{8}$$

$$n=3, \quad a_5 = \frac{-4a_3}{20} = -\frac{1}{5} \left(-\frac{a_1}{3}\right) = \frac{a_1}{15}$$

From equation (11) we get,

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots \\ &= a_0 + a_1 x - \frac{a_0}{2} x^2 - \frac{a_1}{3} x^3 + \frac{a_0}{8} x^4 + \frac{a_1}{15} x^5 + \dots \\ &= a_0 \left( 1 - \frac{x^2}{2} + \frac{x^4}{8} + \dots \right) + a_1 \left( x - \frac{x^3}{3} + \frac{x^5}{15} + \dots \right) \end{aligned}$$

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10.04.2023

# Solve  $y'' + 8xy' - 4y = 0$  by Taylor method.

Solution:-

Given,  $y'' + 8xy' - 4y = 0$

Here,  $P_0(x) = 1$ .

at  $x=0$ ,  $P_0(x) = 1 \neq 0$

Therefore  $x=0$  is an ordinary point.

Consider the series solution

$$y = \sum_{n=0}^{\infty} a_n x^n \quad \text{--- (1)}$$

$$\therefore y' = \sum_{n=0}^{\infty} n a_n x^{n-1} \quad \text{--- (ii)}$$

$$\therefore y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} \quad \text{--- (iii)}$$

Putting the value of  $y, y', y''$  into ①.

$$\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} 8n a_n x^{n-1} - \sum_{n=0}^{\infty} 4a_n x^n = 0$$

$$\Rightarrow (n+1)(n+2) a_{n+2} + n+1$$

$$\Rightarrow \sum_{n=0}^{\infty} [n(n-1) a_n x^{n-2} + 8n a_n x^{n-1} - 4a_n x^n] = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + 8n a_n - 4a_n] x^n = 0$$

Since  $x^n \neq 0$

$$\therefore (n+2)(n+1) a_{n+2} + 4n a_n = 0$$

$\therefore$  Therefore relation of ① is

$$a_{n+2} = \frac{-4n a_n}{(n+2)(n+1)} \quad \text{--- (v)}$$

Putting  $n=0, 1, 2, 3, 4, 5, 6, \dots$  in (v) we get.

$$a_2 = 0$$

$$a_3 = \frac{-4a_1}{6} = -\frac{2a_1}{3}$$

$$a_4 = \frac{-8a_2}{12} = 0$$

$$a_5 = \frac{-12a_3}{20}$$

$$= -\frac{3}{5} \left( -\frac{2a_1}{3} \right) = \frac{2}{5} a_1$$

$$a_6 = 0$$

$$a_7 = \frac{-20a_5}{42} = -\frac{10}{21} \left( \frac{2}{5} a_1 \right) = -\frac{4}{21} a_1$$

-----

Thus series solution is...

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + a_7x^7 + \dots$$

$$= a_0 + a_1x + 0 - \frac{2}{3}a_1x^3 + 0 + \frac{2}{5}a_1x^5 + 0 - \frac{4}{21}a_1x^7 + \dots$$

$$= a_0 + a_1 \left( x - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{4}{21}x^7 + \dots \right)$$

(Ans.)

# Frobenius Method

L-6

06.05.2023

# Solve  $2xy'' - xy' + (x-5)y = 0$  by series method.

Solution:-

$$\text{Given, } 2x^2y'' - xy' + (x-5)y = 0 \quad \text{--- (I)}$$

$$\therefore P_0(x) = 2x^2$$

$$P_0(0) = 2 \cdot 0^2 = 0$$

$$\text{i.e., } P_0(x) = 0 \text{ if } x = 0$$

i.e.,  $x=0$  is a singular point

Let the series solution is

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} \quad \text{--- (II)}$$

$$\text{and } y' = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \quad \text{--- (III)}$$

$$\text{and } y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} \quad \text{--- (IV)}$$

Putting  $y, y'$  and  $y''$  in (I).

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} - x \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} +$$

$$(x-5) \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 2(n+r)(n+r-1) a_n x^{n+r} - \sum_{n=0}^{\infty} (n+r) a_n x^{n+r} +$$

$$\sum_{n=0}^{\infty} a_n x^{n+r+1} - \sum_{n=0}^{\infty} 5a_n x^{n+r} = 0$$

Equating the coefficient of lowest power of  $x$ , we get...

$$2/2 \quad a_0 (n+1) (2n-5) = 0$$

$$\Rightarrow (n+1) (2n-5) = 0 \quad [\because a \neq 0]$$

$$= 2n^2 - 5n + 2n - 5 = 0$$

$$\therefore n = -1, \frac{5}{2}$$

Again in (v) equating the zero the coefficient of highest power of  $x$ , we get...

$$a_{n+1} (n+n+2) (2n+2n-3) + a_n = 0$$

$$\therefore a_{n+1} = \frac{-a_n}{(n+n+2) (2n+2n-3)} \quad \text{--- (iv)}$$

Putting  $n=0, 1, 2, 3, 4, \dots$  in (iv) we get....

$$a_1 = \frac{-a_0}{(n+2) (2n-3)}$$

$$a_2 = \frac{-a_1}{(n+3) (2n-1)} = \frac{a_0}{(n+3) (2n-1) (n+2) (2n-3)}$$

$$a_3 = \frac{-a_2}{(n+4) (2n+1)} \quad \text{and so on.}$$

From ① we get

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$= a_0 x^r + a_1 x^{r+1} + a_2 x^{r+2} + a_3 x^{r+3} + \dots$$

$$\Rightarrow y = x^r (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots)$$

$$= x^r \left[ a_0 - \frac{a_0 x}{(r+2)(2r-3)} + \frac{a_2 x^2}{(r+3)(2r-1)} - \frac{a_3 x^3}{(r+4)(2r+1)} - \dots \right]$$

$$= x^r \left[ a_0 - \frac{a_0 x}{(r+2)(2r-3)} + \frac{a_0 x^2}{(r+3)(2r-1)(r+2)(2r-3)} - \frac{a_0 x^3}{(r+4)(r+1)(r+3)(2r-1)(r+2)(2r-3)} + \dots \right]$$

$$\therefore \frac{y}{a_0} = x^r \left[ 1 - \frac{x}{(r+2)(2r-3)} + \frac{x^2}{(r+3)(2r-1)(r+2)(2r-3)} - \frac{x^3}{(r+4)(2r+1)(r+3)(2r-1)(r+2)(2r-3)} + \dots \right]$$

\* Frobenius Method for series Solution :-

Step-1 : for D.E.  $P_0(x)y'' + P_1(x)y' + P_2(x)y = 0$   
check  $x = x_0$  singular point.

Step 2 : Consider series solution as,  
 $y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots$   
and find  $y'$  and  $y''$ .

Step 3 : Putting  $y, y', y''$  into given equation.  
and find indicial equation for obtain  $m_1, m_2$ .

Step 4 : ① If  $m_1 \neq m_2$  and  $m_1 - m_2 \neq$  an integer  
then complete solution is,

$$y = C_1 (y)_{m=m_1} + C_2 (y)_{m=m_2}$$

② If  $m_1 = m_2$  then complete sol<sup>n</sup>

$$y = C_1 (y)_{m=m_1} + C_2 \left( \frac{\partial y}{\partial m} \right)_{m=m_2}$$

③ If  $m_1 \neq m_2$  but  $m_1 - m_2 =$  an integer  
then complete sol<sup>n</sup> is

$$y = C_1 (y)_{m=m_1} + C_2 \left( \frac{\partial y}{\partial m} \right)_{m=m_2} \quad [m_1 > m_2]$$

# Solve  $2x^2 \left( \frac{d^2y}{dx^2} \right) + x \frac{dy}{dx} - (x+1)y = 0$  by series method / Frobenius method.

Solution :-

(Step 1) Given D.E  $2x^2 \left( \frac{d^2y}{dx^2} \right) + x \frac{dy}{dx} - (x+1)y = 0$  ——— ①

Here,  $P_0(x) = 2x^2$

$\therefore P_0(0) = 2 \cdot 0^2 = 0$

$\therefore x = 0$  is a singular Point.

(Step 2) Consider series solution.

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots$$

$$\therefore \frac{dy}{dx} = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots$$

$$\text{and, } \frac{d^2y}{dx^2} = m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots$$

(Step 3) Putting  $y, \frac{dy}{dx}, \frac{d^2y}{dx^2}$  into ①.

$$\begin{aligned} & 2x^2 \{ m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots \} \\ & + x \{ m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots \} \\ & - (x+1) \cdot (a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots) = 0 \end{aligned}$$

→  $x$  এর সবথেকে ছোট ক্ষমতার সহগগুলোর যোগফল = Indicial Equation

$$\Rightarrow 2 \left[ m(m-1)a_0 x^m + (m+1)ma_1 x^{m+1} + (m+2)(m+1)a_2 x^{m+2} + \dots \right] \\ + \left[ ma_0 x^m + (m+1)a_1 x^{m+1} + m+2 a_2 x^{m+2} + \dots \right] \\ - \left[ a_0 x^{m+1} + a_1 x^{m+2} + a_2 x^{m+3} + \dots \right] - \left[ a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots \right]$$

Indicial equation :-

$$2m(m-1)a_0 + ma_0 - a_0 = 0$$

$$\Rightarrow (2m^2 - 2m + m - 1)a_0 = 0$$

$$\therefore 2m^2 - m - 1 = 0$$

$$[a_0 \neq 0]$$

$$\Rightarrow 2m^2 - 2m + m - 1 = 0$$

$$\Rightarrow 2m(m-1) + 1(m-1) = 0$$

$$\Rightarrow (2m+1)(m-1) = 0$$

$$\therefore m = 1, -\frac{1}{2} \quad \text{i.e., } m_1 = 1, m_2 = -\frac{1}{2}$$

Since  $m_1 \neq m_2$  and  $m_1 - m_2 \neq$  an integer.

Therefore complete solution is.

$$y = C_1(y)_{m=m_1} + C_2(y)_{m=m_2} \quad \text{--- (1)}$$

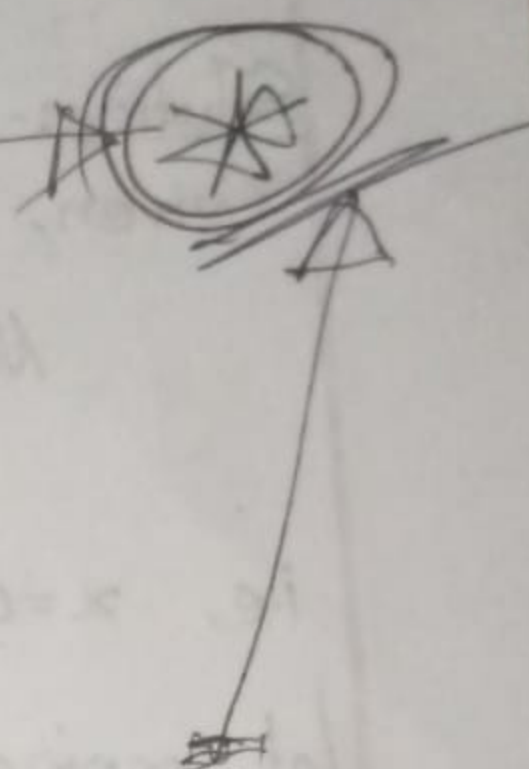
Equating coefficient of  $x^{m+1}$  we get

$$2m(m+1)a_1 + (m+1)a_1 - a_0 - a_1 = 0$$

$$\Rightarrow (2m^2 + 2m + m + 1 - 1)a_1 = a_0$$

$$\Rightarrow (2m^2 + 3m)a_1 = a_0$$

$$\therefore a_1 = \frac{a_0}{2m^2 + 3m}$$



Putting  $a_0$  and  $a_1$  into  $y$ .

$$y = a_0 x^m + \frac{a_0}{2m^2 + 3m} x^{m+1} + \frac{a_0}{(m+1)(2m+5)(2m^2 + 3m)} x^{m+2} + \dots$$

Now,

$$(y)_{m=1} = a_0 x + \frac{a_0}{5} x^2 + \frac{a_0}{20} x^3 + \dots$$

$$\text{and } (y)_{m=-\frac{1}{2}} = a_0 x^{-\frac{1}{2}} + \frac{a_0}{-1} x^{\frac{1}{2}} + \frac{a_0}{-2} x^{\frac{3}{2}} + \dots$$

Therefore complete solution is.

$$y = C_1 \left( a_0 x + \frac{a_0}{5} x^2 + \frac{a_0}{20} x^3 + \dots \right) + C_2 \left( a_0 x^{-\frac{1}{2}} - a_0 x^{\frac{1}{2}} - \frac{a_0}{2} x^{\frac{3}{2}} + \dots \right)$$

(Ans).

[which is the required solution]

# Solve  $x^2 y'' - x y' + (x^2 + 1)y = 0$  by series method.

Solution:-

Given,  $x^2 y'' - x y' + (x^2 + 1)y = 0$  ————— (1)

Here,  $P_0(x) = x^2$

$P_0(0) = 0^2 = 0$

i.e,  $x=0$  is a singular point.

Let series solution is

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots \quad (1)$$

$$y' = a_0 m x^{m-1} + a_1 (m+1) x^m + a_2 (m+2) x^{m+1} + a_3 (m+3) x^{m+2} + \dots \quad (2)$$

$$y'' = a_0 m(m-1) x^{m-2} + a_1 (m+1)m x^{m-1} + a_2 (m+2)(m+1) x^m + a_3 (m+3)(m+2) x^{m+1} + \dots \quad (3)$$

Putting  $y, y', y''$  into (1) we get

$$\begin{aligned} & x^2 \{ m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots \} \\ & - x \{ m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots \} \\ & + (x^2 + 1) \{ a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots \} = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & m(m-1) a_0 x^m + (m+1)m a_1 x^{m+1} + (m+2)(m+1) a_2 x^{m+2} + \dots \\ & - m a_0 x^m - (m+1) a_1 x^{m+1} - (m+2) a_2 x^{m+2} + \dots \\ & + a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots = 0 \end{aligned}$$

∴ Thus series solution is.

$$y = a_0 + a_1 x + a_1 x + \frac{-a_1 - 2a_0}{6} x^3 + \frac{a_1}{6} x^4 + \frac{3a_1 + 6a_0}{40} x^5 + \dots$$

$$= a_0 \left( 1 - \frac{x^3}{3} + \frac{3}{20} x^5 + \dots \right) + a_1 \left( x - \frac{x^3}{6} - \frac{x^4}{6} + \frac{3x^5}{40} + \dots \right)$$

$$\therefore y' = a_0 \left( -x^2 + \frac{3}{4} x^4 + \dots \right) + a_1 \left( 1 - \frac{x^2}{2} - \frac{2x^3}{3} + \dots \right)$$

Now  $y(0) = 1$

i.e.,  $1 = a_0$

and,  $y'(0) = 1$

i.e.,  $1 = a_1$

∴ putting the value  $(a_0, a_1)$  in equation (ii) & (iii)

$$y = 1 - \frac{x^3}{3} + \frac{3}{20} x^5 + \dots + x - \frac{x^3}{6} - \frac{x^4}{6} + \frac{3x^5}{40} + \dots$$

$$= 1 + x - \frac{x^3}{2} - \frac{x^4}{6} + \frac{3x^5}{20}$$

### Practice Set 3

1. If  $P(x)y'' + Q(x)y' + R(x)y = 0$  is differential equation then  $x = x_0$  is a ordinary point if  $P(x) \neq 0$  at  $x_0$ .

2. If  $P(x)y'' + Q(x)y' + R(x)y = 0$  is differential equation then  $x = x_0$  is called ordinary point if  $P(x) \neq 0$  at  $x_0$ .

3. If  $P(x)y'' + Q(x)y' + R(x)y = 0$  is differential equation then  $x = x_0$  is called singular point if  $P(x) = 0$  at  $x = x_0$ .

4. Solve  $y'' + x y' + y = 0$  by series solution.

$$\Rightarrow \text{D.E} = y'' + x y' + y = 0 \quad \text{--- (1)}$$

$$\text{Here, } P(x) = 1 \neq 0$$

Therefore  $x = 0$  is an ordinary point.

Consider a series solution of (1) is...

$$y = \sum_{n=0}^{\infty} a_n x^n \quad \text{--- (2)}$$

$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1} \quad \text{--- (3)}$$

$$y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} \quad \text{--- (4)}$$

Putting the value  $y, y' & y''$  in equ. (1)

$$\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} n a_n x^n + n(n-1) a_n x^{n-2} = 0 \quad \text{--- (2)}$$

From Recurrence formula,

$$(n+2)(n+1) a_{n+2} + n a_n + a_n = 0$$

$$\Rightarrow a_{n+2} = \frac{-(n a_n + a_n)}{(n+2)(n+1)} \quad \text{--- (3)}$$

Putting the value  $0, 1, 2, 3, \dots$  in eq. (3)

$$a_2 = \frac{-a_0}{2} = -\frac{a_0}{2}$$

$$a_3 = \frac{-2a_1}{6} = -\frac{a_1}{3}$$

$$a_4 = \frac{-3a_2}{12} = \frac{a_0}{8}$$

From equation (1) putting  $a_2 + a_3 + a_4$ .

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$= a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$= a_0 + a_1 x - \frac{a_0}{2} x^2 - \frac{a_1}{3} x^3 + \frac{a_0}{8} x^4 + \dots$$

# Solve.  $y'' + 8xy' - 4y = 0$  by power method.

$\Rightarrow$  Cr. D.E  $\Rightarrow y'' + 8xy' - 4y = 0$  ———— (1)

Here,  $P_0(x) = 1 \neq 0$

Therefore, ~~at  $x=0$~~   $x=0$  is an ordinary point.

So,  $y = \sum_{n=0}^{\infty} a_n x^n$  ———— (2)

$y' = \sum_{n=0}^{\infty} n a_n x^{n-1}$  ———— (3)

$y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$  ———— (4)

Putting the value  $y, y', y''$  in eq. (1):

$$\sum_{n=0}^{\infty} [n(n-1) a_n x^{n-2} + 8n a_n x^{n-1} - 4a_n x^n] = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} x^n + 8(n+1) a_{n+1} x^n - 4a_n x^n] = 0$$

$$= \sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + 8(n+1) a_{n+1} - 4a_n] x^n = 0$$

Since  $x^n \neq 0$ ,

So,  $(n+2)(n+1) a_{n+2} + 8(n+1) a_{n+1} - 4a_n = 0$

$$\therefore a_{n+2} = \frac{4a_n - 8(n+1)a_{n+1}}{(n+1)(n+2)} \quad \text{--- (5)}$$

Putting  $0, 1, 2, 3, 4, \dots$  in (v) all other

$$a_2 = a$$

$$a_3 = -\frac{2a_1}{3}$$

$$a_4 = 0$$

The value of  $a_2, a_3, a_4, \dots$  in equ. (iv)

$$f = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \dots$$

$$= a_0 + a_1 x + 0 + \frac{2a_1}{3} x^2 + 0 + \dots$$

$$1 + v + v^2 + v^3 + \dots = (v) f$$

$$1 + 0 + 0 + 0 + \dots = (0) f$$

$$0 \neq 1 =$$

(ii) To find  $y$  consider  $x=0$  is

Consider the series solution

$$\dots + v^n + \dots + v^2 + v + 1 = f$$

$$\dots + v^n + \dots + v^2 + v + 1 = f$$

$$\dots + v^n + \dots + v^2 + v + 1 = f$$

# Solve the following series in powers of  $x-1$   
 $x^2 y'' + x y' + y = 0$

Solution:

Given,  $x^2 y'' + x y' + y = 0$  ————— ①

Let,  $v = x-1$

$\Rightarrow v+1 = x$

$\therefore ① \Rightarrow (v+1)y'' + (v+1)y' + y = 0$  ————— ②

$\Rightarrow (v^2 + 2v + 1)y'' + (v+1)y' + y = 0$

$P(v) = v^2 + 2v + 1$

$\therefore P_0(0) = 0^2 + 2 \cdot 0 + 1$   
 $= 1 \neq 0$

i.e.,  $v=0$  is an ordinary point of ②.

Consider the series solution.

$y = a_0 + a_1 v + a_2 v^2 + a_3 v^3 + \dots + a_n v^n + \dots$

$\therefore y' = a_1 + 2a_2 v + 3a_3 v^2 + \dots + na_n v^{n-1} + \dots$

$\therefore y'' = 2a_2 + 6a_3 v + \dots + n(n-1)a_n v^{n-2} + \dots$

Putting the value of  $y, y', y''$  into (ii).

$$\begin{aligned}
 & (b^2 + 2bv + 1) \{ 2a_2 + 6a_3v + \dots + n(n-1)a_nv^{n-2} + \dots \} \\
 & + (b+1) (a_1 + 2a_2v + 3a_3v^2 + \dots + na_nv^{n-1} + \dots) \\
 & + a_0 + a_1v + a_2v^2 + a_3v^3 + \dots + a_nv^n + \dots = 0.
 \end{aligned}$$

Equating the coefficient of  $v^0$ , we get...

$$2a_2 + a_1 + a_0 = 0.$$

$$\therefore 2a_2 = -a_1 - a_0$$

Equating coefficient of  $v$ , we get...

$$4a_2 + 6a_3 + a_1 + 2a_2 + a_1 = 0$$

$$\Rightarrow 2a_1 + 2a_2 + 6a_3 = 0.$$

$$\Rightarrow a_3 = \frac{-2a_2 - 2a_1}{6}$$

$$= \frac{-b \cdot \frac{-a_1 - a_0}{2} - 2a_1}{6}$$

$$= \frac{3a_1 + 3a_0 - 2a_1}{6}$$

$$= \frac{a_1 + 3a_0}{6}$$

Recurrence formula is .

$$n(n-1)a_n + 2n(n+1)a_{n+1} + (n+2)(n+1)a_{n+2} + na_n(n+1)a_{n+1} + a_n = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} + (2n^2 + 2n + n + 1)a_{n+1} + (n^2 - n + n + 1)a_n = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} + (2n^2 + 3n + 1)a_{n+1} + (n^2 + 1)a_n = 0$$

$$\Rightarrow a_{n+2} = - \frac{(2n^2 + 3n + 1)a_{n+1} + (n^2 + 1)a_n}{(n+2)(n+1)}$$

Putting  $n=2, 3, \dots$

$$\underline{n=2} \quad a_4 = - \frac{(8+6+1)a_3 + 5a_2}{12}$$

$$= - \frac{15a_3 + 5a_2}{12}$$

$$= - \frac{15 \frac{a_1 + 3a_0}{6} + 5 \frac{-a_1 - a_0}{2}}{12}$$

$$= - \frac{5a_1 + 15a_0 - 5a_1 - 5a_0}{24}$$

$$= - \frac{10a_0}{24}$$

$$= - \frac{5a_0}{12}$$

$$\begin{aligned}
 \text{and } a_5 &= - \frac{27a_4 + 10a_3}{20} \\
 &= - \frac{27 \left( -\frac{5a_0}{12} + 10 \frac{a_1 + 3a_0}{6} \right)}{20} \\
 &= - \frac{-\frac{45a_0}{4} + \frac{5a_1 + 15a_0}{3}}{20} \\
 &= - \frac{-135a_0 + 20a_1 + 60a_0}{240} \\
 &= - \frac{-75a_0 + 20a_1}{240} = \frac{15a_0 - 4a_1}{48}
 \end{aligned}$$

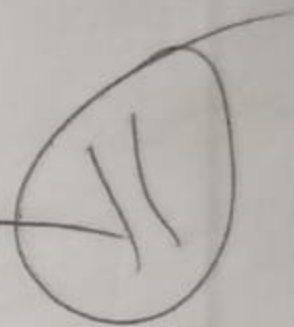
Putting  $a_2, a_3, a_4, a_5$  ... into (iii),

$$\begin{aligned}
 y &= a_0 + a_1 v + \frac{-a_1 - a_0}{2} v^2 + \frac{a_1 + 3a_0}{6} v^3 + \frac{-5a_0}{12} v^4 + \dots \\
 &= a_0 \left( 1 - \frac{v^2}{2} + \frac{v^3}{2} - \frac{5v^4}{12} \dots \right) + a_1 \left( v - \frac{v^2}{2} + \frac{v^3}{6} + \dots \right) \\
 &= a_0 \left\{ 1 - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{2} - \frac{5(x-1)^4}{12} \right\} + a_1 \left\{ (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{2} + \dots \right\}
 \end{aligned}$$

Lecture 11

Practice# Solve  $y'' + xy' + y = 0$  by Taylor method.Solution:-Given D.E  $\Rightarrow y'' + xy' + y = 0$  ——— ①Here  $P_0(x) = 1$ , i.e.,  $x=0$  is an ordinary point.  
Consider a series solution of ①.

$$y = \sum_{n=0}^{\infty} \cancel{a_n x^n} a_n x^n$$



$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$$

Putting the value of  $y, y'$  &  $y''$  in equation ①.

$$\sum_{n=0}^{\infty} \{ n(n-1) a_n x^{n-2} + n a_n x^n + a_n x^n \} = 0$$

By Recurrence formula,

$$(n+2)(n+1) a_{n+2} + n a_n + a_n = 0$$

$$\Rightarrow a_{n+2} = -\frac{n a_n + a_n}{(n+1)(n+2)}$$

putting  $n=0$ ,  $a_2 = -\frac{0 \cdot a_0 + a_0}{2} = -\frac{a_0}{2}$  ✓

$n=1$ ,  $a_3 = -\frac{a_1 - a_1}{6} = -\frac{a_1}{3}$  ✓

$n=2$ ,  $a_4 = \frac{-3a_2}{12} = -\left(\frac{3}{12}\right) \left(-\frac{a_0}{2}\right)$   
 $= \frac{a_0}{8}$  ✓

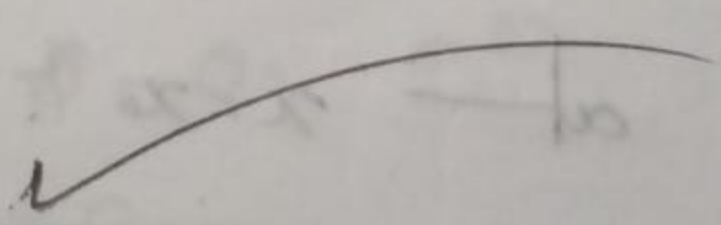
$n=3$ ,  $a_5 = \frac{-4a_3}{20} = \left(-\frac{4}{20}\right) \left(-\frac{a_1}{3}\right)$   
 $= \frac{a_1}{15}$  ✓

From equation (1), we get...

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$= a_0 + a_1 x - \frac{a_0}{2} x^2 - \frac{a_1}{3} x^3 + \frac{a_0}{8} x^4 + \frac{a_1}{15} x^5 + \dots$$



Q Type-2:  $(x^2+1)y'' + xy' + 2xy = 0$ ,  $y(0) = 1$ ,  $y'(0) = 1$   
by Taylor

## Ordinary Point :-

Type 01:  $y'' + xy' + y = 0$  by Taylor method.

Type 02:  $(x^2+1)y'' + xy' + 2xy = 0$ ,  $y(0) = 1$ ,  $y'(0) = 1$ .  
by Taylor method.

Type-03:  $x^2y'' + xy' + y = 0$  in power of  $(x-1)$ .

If  $P_0(x)y'' + P_1(x)y' + P_2(x)y = 0$  is a DE.  
then  $x = x_0$  is called Ordinary point if  
 $P_0(x) \neq 0$  at  $x = x_0$ .

If  $P_0(x)y'' + P_1(x)y' + P_2(x)y = 0$  is DE then  
 $x = x_0$  is called Singular Point if  
 $P_0(x) = 0$  at  $x = x_0$ .

$$\textcircled{2} \quad 2x^2 \left( \frac{d^2 y}{dx^2} \right) + x \frac{dy}{dx} - (x+1)y = 0 \quad \text{--- (1)}$$

Here  $P_0(x) = 2x^2$   
 $= 2 \cdot 0$   
 $= 0$

$x=0$  is a Singular point.

Now,

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$$

$$y' = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots$$

$$y'' = m(m-1) a_0 x^{m-2} + m(m+1) a_1 x^{m-1} + (m+1)(m+2) a_2 x^m + \dots$$

into (1),

$$2m(m-1) a_0 x^m + 2m(m+1) a_1 x^{m+1} + 2(m+1)(m+2) a_2 x^{m+2} + \dots$$

$$+ m a_0 x^m + (m+1) a_1 x^{m+1} + (m+2) a_2 x^{m+2} + \dots$$

$$= a_0 x^{m+1} + a_1 x^{m+2} + a_2 x^{m+3} + \dots - a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$$

By,

Indicial Formula,

$$2m(m-1) a_0 + m a_0 - a_0 = 0$$

$$(2m^2 - 2m + m - 1) = 0$$

$$2m(m-1) + 1(m-1) = 0$$

$$\therefore m = 1, -\frac{1}{2}$$

$m_1 \neq m_2$ ,  $m_1 > m_2$  not integer

## Singular Point :-

Type 01 :  $2x^2y'' - xy' + (x-5)y = 0$

Type 02 :  $2x^2y'' + xy' - (x+1)y = 0$

Type 03 :  $x^2y'' - xy' + (x^2+1)y = 0$

①  $m_1 \neq m_2$  and  $m_1 - m_2 \neq \text{integer}$ .

then,  $y = C_1 (y)_{m=m_1} + C_2 (y)_{m=m_2}$

②  $m_1 = m_2$

then,  $y = C_1 (y)_{m=m_1} + C_2 \left( \frac{\partial y}{\partial x} \right)_{m=m_1}$

③  $m_1 \neq m_2$  and  $m_1 - m_2 = \text{integer}$ .

then,  $y = C_1 (y)_{m=m_1} + \left( \frac{\partial y}{\partial m} \right)_{m=m_2}$

Q Solve.  $2x^2y'' + xy' + (x-5)y = 0$

Solution :

Given D.E =  $2x^2y'' + xy' + (x-5)y = 0$  ——— ①

Here,  $P_0(x) = 2x^2$

$P_0(0) = 2 \cdot 0^2 = 0$

i.e,  $x=0$  is a singular number.

Now,

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots$$

$$y' = m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + (m+3) a_3 x^{m+2} + \dots$$

$$y'' = m(m-1) a_0 x^{m-2} + m(m+1) a_1 x^{m-1} + (m+1)(m+2) a_2 x^m + \dots$$

Then the value of  $y$ ,  $y'$  &  $y''$  putting into ①.

$$2x^2 [m(m-1) a_0 x^{m-2} + m(m+1) a_1 x^{m-1} + (m+1)(m+2) a_2 x^m + \dots]$$

$$+ x [m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + (m+3) a_3 x^{m+2} + \dots]$$

$$+ (x-5) [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots]$$

By Indicial Equation,

$$2m(m+1)a_0 - ma_0 - 5a_0 = 0$$

$$\Rightarrow (2m^2 - 2m - m - 5)a_0 = 0$$

$$\Rightarrow 2m^2 - 3m - 5 = 0 \quad [\because a_0 \neq 0]$$

$$\Rightarrow 2m^2 - 5m + 2m - 5 = 0$$

$$\Rightarrow 2m^2 + 2m - 5m - 5 = 0$$

$$\Rightarrow 2m(m+1) - 5(m+1) = 0$$

$$\Rightarrow (m+1)(2m-5) = 0$$

$$\therefore m = -1, \frac{5}{2}$$

Since  $m_1 \neq m_2$  and  $m_1 - m_2 \neq \text{integer}$ .

So the complete solution is...

$$y = C_1(y)_{m=m_1} + C_2(y)_{m=m_2}.$$

Equating the coefficient of  $m+1$ , then:

$$2m(m+1)a_1 - (m+1)a_1 + a_0 - 5a_1 = 0$$

$$\Rightarrow (2m^2 + 2m - m - 1 - 5)a_1 = -a_0$$

$$\therefore a_1 = \frac{-a_0}{(2m^2 + m - 6)}.$$

Equating the coefficient of  $m+2$ , then,

$$2(m+1)(m+2)a_2 - (m+2)a_2 + a_1 - 5a_2 = 0$$

$$\Rightarrow (2m^2 + 4m + 2m + 4 - m - 2 - 5)a_2 = -a_1$$

$$\Rightarrow a_2 = \frac{-a_1}{2m^2 + 5m - 3}$$

putting  $a_1, a_2$  into  $y$ .

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$$

$$= a_0 x^m + \frac{a_0}{(2m^2 + m - 6)} x^{m+1} - \frac{a_1}{(2m^2 + 5m - 3)} x^{m+2} + \dots$$

Now,

$$(y)_{m=-1} \Rightarrow a_0 x^{-1} - \frac{a_0}{5} - \frac{a_0 x}{30} + \dots$$

$$(y)_{m=\frac{5}{2}} \Rightarrow a_0 x^{\frac{5}{2}} - \frac{a_0 \cdot x^{\frac{7}{2}}}{9} - \frac{a_0 x^{\frac{9}{2}}}{\frac{47}{2}} + \dots$$

$$= a_0 x^{\frac{5}{2}} - \frac{a_0 x^{\frac{7}{2}}}{9} - \frac{2a_0 x^{\frac{9}{2}}}{47} + \dots$$

# Solution of Laplace equation:

Laplace Equation is  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$  ————— ①

Consider the required solution of ① by separation of variable method is ————— ②

$$u(x, y) = X(x) \cdot Y(y)$$

Differentiating partially with respect to  $x$  and  $y$ , we get —

$$\frac{\partial^2 u}{\partial x^2} = Y \frac{d^2 X}{dx^2}$$

and  $\frac{\partial^2 u}{\partial y^2} = X \frac{d^2 Y}{dy^2}$  ————— ③

using ③ in ①, we get —

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0$$

$$\Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = - \frac{1}{Y} \frac{d^2 Y}{dy^2}$$
 ————— ④

Since R.S. of ④ is a function of  $x$  only and L.S. of ④ is a function of  $y$  only, so both sides of ④ are equal to a constant say  $k$ .

$$\text{i.e., } \frac{1}{x} \frac{d^2 x}{dx^2} = -\frac{1}{y} \frac{d^2 y}{dy^2} = k$$

$$\Rightarrow \frac{d^2 x}{dx^2} - kx = 0 \quad \text{--- (v)}$$

$$\text{and } \frac{d^2 y}{dy^2} + ky = 0 \quad \text{--- (vi)}$$

Equation (v) & (vi) are ~~separate~~ ordinary differential equations.

Case - 1 : if  $k = \lambda^2 > 0$  then equation (v) & (vi) are

$$\frac{d^2 x}{dx^2} - \lambda^2 x = 0$$

$$\frac{d^2 y}{dy^2} + \lambda^2 y = 0$$

which can be written as  $(D^2 - \lambda^2)x = 0$

$$(D^2 + \lambda^2)y = 0$$

Auxiliary equations are

$$m^2 - \lambda^2 = 0$$

$$\Rightarrow m = \pm \lambda$$

$$\text{and } m^2 = -\lambda^2 \Rightarrow m = \pm i\lambda$$

Then the solution of these equation are

$$x = Ae^{\lambda x} + Be^{-\lambda x}$$

$$\text{and } y = C \cos \lambda y + D \sin \lambda y$$

Substituting  $X$  and  $Y$  in equation (i), we get...

$$V(x, y) = (A e^{\lambda x} + B e^{-\lambda x}) \cdot (C \cos \lambda y + D \sin \lambda y).$$

case II:

If  $K = -\lambda^2 \leq 0$  then equation (v) & (vi) are

$$\frac{d^2 X}{dx^2} + \lambda^2 X = 0$$

$$\frac{d^2 Y}{dy^2} - \lambda^2 Y = 0$$

which can be written as

$$(D^2 + \lambda^2) X = 0$$

$$(D^2 - \lambda^2) Y = 0$$

Auxiliary equations are,

$$m^2 + \lambda^2 = 0$$

$$\Rightarrow m = \pm i\lambda$$

and  $m^2 - \lambda^2 = 0$

$$\therefore m = \pm \lambda$$

Then the solution of those equations are

$$X = C \cos \lambda x + D \sin \lambda x$$

$$Y = A e^{\lambda y} + B e^{-\lambda y}$$

substituting  $X$  and  $Y$  in (i), we get.

$$V(x, y) = (C \cos \lambda x + D \sin \lambda x) \cdot (A e^{\lambda y} + B e^{-\lambda y})$$

15.07.2023

Laplace Transform

Let  $F(t)$  be a function of  $t$  defined for all  $t \geq 0$ , then the Laplace transform of  $F(t)$  denoted by  $\mathcal{L}\{F(t)\}$  or  $f(s)$  is def. defined by  $\mathcal{L}\{F(t)\} = f(s) = \int_0^{\infty} e^{-st} F(t) dt$ .

Problem: Find the Laplace transform of  $t^2$ .

Solution:-

$$\mathcal{L}(t) = \int_0^{\infty} e^{-st} F(t) dt$$

$$= \int_0^{\infty} e^{-st} t dt$$

$$= t \int_0^{\infty} e^{-st} dt - \int_0^{\infty} \left\{ \frac{d}{dt} (t) \int_0^{\infty} e^{-st} dt \right\} dt$$

$$= \left[ \frac{t e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} dt$$

$$= \frac{1}{s} (0 - 0) + \frac{1}{s} \left[ \frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= -\frac{1}{s^2} (e^{-\infty} - e^{-0})$$

$$= -\frac{1}{s^2} (0 - 1)$$

$$= \frac{1}{s^2}$$

(Ans)

$$\left[ \frac{1}{\infty} = 0 \right]$$

Q Problem : Find the Laplace transform of 1.

Solution :-

$$L(1) = \int_0^{\infty} e^{-st} F(1) dt.$$

$$= \int_0^{\infty} e^{-st} \cdot 1 dt.$$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \left[ \frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= \frac{e^{-\infty}}{-s} - \frac{e^{-0}}{-s}$$

$$= 0 + \frac{1}{s}$$

$$= \frac{1}{s} \quad \underline{\underline{Ans.}}$$

Q Problem : Find the Laplace transform of  $e^{at}$ .

Solution :-

$$L(e^{at}) = \int_0^{\infty} e^{-st} F(e^{at}) dt$$

$$= \int_0^{\infty} e^{-st} \cdot e^{at} dt.$$

$$= \int_0^{\infty} e^{-st+at} dt.$$

$$= \int_0^{\infty} e^{-t(s-a)} dt$$

$$= \left[ \frac{e^{-t(s-a)}}{s-a} \right]_0^{\infty}$$

$$= \frac{1}{s-a}$$

Ans.

Q Find the Laplace Transform of the function  $F(t)$ ,  
 where  $f(t) = \begin{cases} t, & 0 < t < 2 \\ 3, & t > 2 \end{cases}$ .

Solution:

$$\text{Given } F(t) = \begin{cases} t, & 0 < t < 2 \\ 3, & t > 2 \end{cases}$$

$$\therefore \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt.$$

$$= \int_0^2 e^{-st} F(t) dt + \int_2^{\infty} e^{-st} F(t) dt.$$

$$= \int_0^2 e^{-st} t dt + \int_2^{\infty} e^{-st} 3 dt.$$

$$= t \int_0^2 e^{-st} dt - \int_0^2 \left\{ \frac{d}{dt} (t) \int_0^2 e^{-st} dt \right\} dt + 3 \int_2^{\infty} e^{-st} dt$$

$$= \left[ \frac{t e^{-st}}{-s} \right]_0^2 - \int_0^2 \frac{e^{-st}}{-s} dt + 3 \cdot \left[ \frac{e^{-st}}{-s} \right]_2^{\infty}$$

$$= \frac{2e^{-2s}}{-s} - 0 + \frac{1}{s} \left[ \frac{e^{-st}}{-s} \right]_0^2 + 3 \left( 0 - \frac{e^{-2s}}{-s} \right)$$

$$= -\frac{2e^{-2s}}{s} + \frac{1}{s} \left( \frac{e^{-2s}}{-s} - \frac{1}{-s} \right) + \frac{3e^{-2s}}{s}$$

$$= \frac{e^{-2s}}{s} - \frac{e^{-2s}}{s^2} + \frac{1}{s^2} \quad (\text{Ans}).$$

## Linearity Property of Laplace Transform:

If  $\lambda_1$  and  $\lambda_2$  are any constants and  $F_1(t)$ ,  $F_2(t)$  are functions with Laplace transform  $\mathcal{L}\{F_1(t)\}$  and  $\mathcal{L}\{F_2(t)\}$  then  ~~$\mathcal{L}\{\lambda_1 F_1(t)\}$~~ .

$$\mathcal{L}\{\lambda_1 F_1(t) + \lambda_2 F_2(t)\} = \lambda_1 \mathcal{L}\{F_1(t)\} + \lambda_2 \mathcal{L}\{F_2(t)\}.$$

Proof:

By the definition of L.T.

$$\mathcal{L}\{F_1(t)\} = \int_0^{\infty} e^{-st} F_1(t) dt \quad \text{--- (i)}$$

$$\text{and } \mathcal{L}\{F_2(t)\} = \int_0^{\infty} e^{-st} F_2(t) dt \quad \text{--- (ii)}$$

Now,

$$\mathcal{L}\{\lambda_1 F_1(t) + \lambda_2 F_2(t)\} = \int_0^{\infty} e^{-st} \{\lambda_1 F_1(t) + \lambda_2 F_2(t)\} dt$$

$$= \int_0^{\infty} e^{-st} \cdot \lambda_1 F_1(t) dt + \int_0^{\infty} e^{-st} \cdot \lambda_2 F_2(t) dt$$

$$= \lambda_1 \int_0^{\infty} e^{-st} F_1(t) dt + \lambda_2 \int_0^{\infty} e^{-st} F_2(t) dt$$

$$= \lambda_1 \mathcal{L}\{F_1(t)\} + \lambda_2 \mathcal{L}\{F_2(t)\}$$

[Hence Proved]

Q Find L.T. of  $e^{4t} + 4t^3 - 2\sin 3t + 3\cos 5t$ .

Solution :-

$$\text{let } F(t) = e^{4t} + 4t^3 - 2\sin 3t + 3\cos 5t$$

using linearity Property of L.T

$$\mathcal{L}\{F(t)\} = \mathcal{L}\{e^{4t}\} + 4\mathcal{L}\{t^3\} - 2\mathcal{L}\{\sin 3t\} + 3\mathcal{L}\{\cos 5t\}$$

$$= \frac{1}{s-4} + 4 \frac{3!}{s^{3+1}} - 2 \frac{3}{s^2+3^2} + 3 \frac{s}{s^2+5^2}$$

$$= \frac{1}{s-4} + \frac{24}{s^4} - \frac{6}{s^2+9} + \frac{3s}{s^2+25}$$

Ans

Q

First Shifting property / First translation property:

statement : If  $\mathcal{L}\{F(t)\} = f(s)$  then  $\mathcal{L}\{e^{at} F(t)\} = f(s-a)$

Proof :

By the definition of L.T.

$$f(s) = \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$$

$$\begin{aligned}
 \text{Now, } \mathcal{L}\{e^{at}F(t)\} &= \int_0^{\infty} e^{-st} e^{at} F(t) dt \\
 &= \int_0^{\infty} e^{-st+at} F(t) dt \\
 &= \int_0^{\infty} e^{-(s-a)t} F(t) dt \\
 &= f(s-a)
 \end{aligned}$$

Q Find the L.T of  $e^{3t}(2\cos 5t - 3\sin 5t)$

Solution:

$$\begin{aligned}
 \text{Here } f(s) &= \mathcal{L}\{2\cos 5t - 3\sin 5t\} \\
 &= 2\mathcal{L}\{\cos 5t\} - 3\mathcal{L}\{\sin 5t\} \\
 &= 2 \cdot \frac{s}{s^2+5^2} - 3 \cdot \frac{5}{s^2+5^2} \\
 &= \frac{2s}{s^2+25} - \frac{15}{s^2+25}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \mathcal{L}\{e^{3t}(2\cos 5t - 3\sin 5t)\} &= f(s-3) \\
 &= \frac{2(s-3)}{(s-3)^2+25} - \frac{15}{(s-3)^2+25} \\
 &= \frac{2s-6}{s^2-6s+34} - \frac{15}{s^2-6s+34} \\
 &= \frac{2s-21}{s^2-6s+34}
 \end{aligned}$$

Ans

## # Laplace Transform of derivatives :-

\* If  $\mathcal{L}\{F(t)\} = f(s)$  then  $\mathcal{L}\{F'(t)\} = sf(s) - F(0)$

Proof :-

By the definition of L.T.

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt = f(s)$$

$$\therefore \mathcal{L}\{F'(t)\} = \int_0^{\infty} e^{-st} F'(t) dt$$

$$= e^{-st} \int_0^{\infty} F'(t) dt - \int_0^{\infty} \left\{ \frac{d}{dt} (e^{-st}) \right\} F'(t) dt$$

$$= [e^{-st} F(t)]_0^{\infty} - \int_0^{\infty} e^{-st} \cdot (-s) F(t) dt$$

$$= 0 - F(0) + s \int_0^{\infty} e^{-st} F(t) dt$$

$$= sf(s) - F_0$$

[Proved]

# If  $\mathcal{L}\{F(t)\} = f(s)$  then prove that  $\mathcal{L}\{F''(t)\} = s^2 f(s)$

Proof:

By the definition of Laplace transform

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt = f(s)$$

$$\begin{aligned}\therefore \mathcal{L}\{F''(t)\} &= \int_0^{\infty} e^{-st} F''(t) dt \\ &= e^{-st} \int_0^{\infty} F''(t) dt - \int_0^{\infty} \left( \frac{d}{dt} (e^{-st}) \right) \int_0^{\infty} F''(t) dt dt\end{aligned}$$

$$= \left[ e^{-st} F'(t) \right]_0^{\infty} + s \int_0^{\infty} e^{-st} F'(t) dt.$$

$$= 0 - F'(0) + s \{ s f(s) - F(0) \}$$

$$= s^2 f(s) - s F(0) - F'(0)$$

[Proved]

# Find the Laplace transform of  $\cos at$  using theorem

Solution:-

$$\text{Let } F(t) = \cos at$$

$$\text{Then, } \begin{cases} F'(t) = -a \sin at \\ F''(t) = -a^2 \cos at \end{cases} \quad \begin{cases} F(0) = 1 \\ F'(0) = 0 \end{cases}$$

Now by theorem II, we have

$$\mathcal{L}\{F'(t)\} = s^2 \mathcal{L}\{F(t)\} - sF(0) - F'(0).$$

$$\mathcal{L}\{-a^2 \cos at\} = s^2 \mathcal{L}\{\cos at\} - s \cdot 1 - 0.$$

$$\text{Or, } -a^2 \mathcal{L}\{\cos at\} = s^2 \mathcal{L}\{\cos at\} - s.$$

$$\text{Or, } (s^2 + a^2) \mathcal{L}\{\cos at\} = s$$

$$\text{Or, } \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

# Find the L.T. of  $t^2 \cos at$  using theorem 11.

Solution:-

$$\text{Since } \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}, \quad s > 0$$

$$\therefore \mathcal{L}\{t^2 \cos at\} = (-1)^2 \frac{d^2}{ds^2} \mathcal{L}\{\cos at\}$$

$$= \frac{d^2}{ds^2} \left\{ \frac{s}{s^2 + a^2} \right\}$$

$$= \frac{d}{ds} \left\{ \frac{-2s^2}{(s^2 + a^2)^2} + \frac{1}{s^2 + a^2} \right\}$$

$$= \frac{8s^3}{(s^2 + a^2)^3} - \frac{6s}{(s^2 + a^2)^2}$$

$$= \frac{8s^3 - 6s^3 - 6a^2s}{(s^2 + a^2)^3}$$

$$= \frac{2s(s^2 - 3a^2)}{(s^2 + a^2)^3}$$

# Find the Laplace transform of  $\int_0^t \frac{\sin t}{4} dt$ .

Solution:-

We know that  $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1} = f(s)$  (say).

$$\text{Now, } \mathcal{L}\left\{\frac{\sin t}{4}\right\} = \int_0^\infty F(u) du = \int_0^\infty \frac{1}{u^2+1} du.$$

$$= \left[ \tan^{-1} u \right]_0^\infty = \tan^{-1} \infty - \tan^{-1} 0$$

$$= \frac{\pi}{2} - \tan^{-1} 0 = \cot^{-1} 0 = \tan^{-1} \frac{1}{0}$$

$$\text{Since } \tan^{-1} \infty + \cot^{-1} 0 = \frac{\pi}{2}.$$

$$\text{Now we also know that } \mathcal{L}\left\{\int_0^t F(u) du\right\} = \frac{f(s)}{s}.$$

$$\therefore \mathcal{L}\left\{\int_0^t \frac{\sin t}{4} dt\right\} = \frac{1}{s} f(s) = \frac{1}{s} \tan^{-1} \frac{1}{s}$$

Ans.

Theorem 8.1

If  $\mathcal{L}\{f(t)\} = f(s)$  then  $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} f(s)$ .

~~12.10.20~~

Q Prove that.  $\int_0^\infty t e^{-st} \sin at \cdot dt = \frac{2as}{(s^2+a^2)^2}$

$$L.S = \int_0^\infty e^{-st} t \sin at \cdot dt.$$

$$= \mathcal{L}\{t \sin at\}$$

$$= (-1)^1 \frac{d}{ds} \mathcal{L}\left\{\frac{a}{s^2+a^2}\right\}$$

$$= - (s^2+a^2) \frac{d}{ds} a - a \cdot \frac{d}{ds} (s^2+a^2)$$

$$= \frac{2as}{(s^2+a^2)^2}$$

$$= \frac{2as}{(s^2+a^2)^2}$$

[Hence Proved]

Q Find  $\mathcal{L}\{t^2(\sin at + e^{at})\}$

Solutions :-

$$\mathcal{L}\{t^2(\sin at + e^{at})\}$$

$$= \mathcal{L}\{t^2 \sin at\} + \mathcal{L}\{t^2 e^{at}\}$$

$$= (-1)^2 \frac{d^2}{ds^2} \frac{a}{s^2 + a^2} + (-1)^2 \frac{d^2}{ds^2} \frac{1}{s-a}$$

$$= \frac{d^2}{ds^2} \left( \frac{a}{s^2 + a^2} + \frac{1}{s-a} \right)$$

$$= \frac{d}{ds} \left\{ \frac{-2as}{(s^2 + a^2)^2} - \frac{1}{(s-a)^2} \right\}$$

$$= \frac{(s^2 + a^2)^2(-2a) + 2as \cdot 2(s^2 + a^2)2s}{(s^2 + a^2)^4} + \frac{2}{(s-a)^3}$$

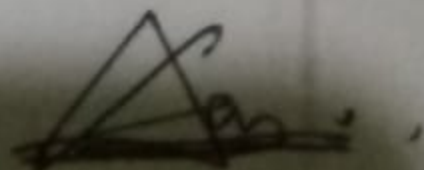
$$= \frac{-2a(s^2 + a^2) + 8as^2}{(s^2 + a^2)^3} + \frac{2}{(s-a)^3}$$

$$= \frac{-2as^2 - 2a^3 + 8as^2}{(s^2 + a^2)^3} + \frac{2}{(s-a)^3}$$

$$= \frac{-2as^2 - 2a^3 + 8as^2}{(s^2 + a^2)^3} + \frac{2}{(s-a)^3}$$

$$= \frac{6as^2 - 2a^3}{(s^2 + a^2)^3} + \frac{2}{(s-a)^3}$$

$$= \frac{2a(3s^2 - a^2)}{(s^2 + a^2)^3} + \frac{2}{(s-a)^3}$$



# Solve  $2xz - px^2 - 2qxy + pq = 0$  by charpit's method.

Solution :-

Given,  $2xz - px^2 - 2qxy + pq = 0$  ————— ①

$\therefore f(x, y, z, p, q) = 2xz - px^2 - 2qxy + pq$ .

Now,  $\frac{\partial f}{\partial x} = 2z - 2px - 2qy$ .

$\frac{\partial f}{\partial y} = -2qx$ .

$\frac{\partial f}{\partial z} = 2x$

$\frac{\partial f}{\partial p} = -x^2 + q$ .

$\frac{\partial f}{\partial q} = -2xy + p$ .

From charpit auxiliary equations, we get...

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-\frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

$$\Rightarrow \frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dz}{px^2 + 2xyq - 2pq} = \frac{dx}{x^3 - q} = \frac{dy}{2xy - p} = \frac{dF}{0}$$

From second, we get..

$$dq = 0$$

Integrating we get...

$$\int dq = 0$$

$$\Rightarrow q = C$$

[where  $C$  is an arbitrary constant]

putting  $q = C$  in (1), we get...

$$2xz - px^2 - 2cxy + pc = 0$$

$$p(c - x^2) = 2cxy - 2xz$$

$$\therefore p = \frac{2x(cy - z)}{c - x^2}$$

$$= \frac{2x(z - cy)}{x^2 - c}$$

From,  $dz = p dx + q dy$ .

$$\text{DE} = 2x^2 \left( \frac{dy}{dx} \right)$$

DE

Formation of DE :-

# Form the DE corresponding to the family of curves  $y = c(x-c)$  where  $c$  is an arbitrary constant.

① differentiate  $\frac{dy}{dx} / D/y'/D$ .

② no constant.

# Given curve  $y = c(x-c)^2$  ——— ①

Differentiating ① w.r.to  $x$ , we get...

$$y' = \frac{dy}{dx} = 2c(x-c) \text{ ——— ②}$$

Dividing ① by ②, we get

$$\frac{y}{y'} = \frac{c(x-c)^2}{2c(x-c)}$$

$$\Rightarrow \frac{y}{y'} = \frac{1}{2}(x-c)$$

$$\Rightarrow x-c = \frac{2y}{y'}$$

$$\Rightarrow c = x - \frac{2y}{y'}$$

putting this value of  $c$  in (a) we get...

$$y' = 2\left(x - \frac{2y}{y'}\right) \left(x - x + \frac{2y}{y'}\right)$$

$$\Rightarrow y' = 2\left(-\frac{xy' - 2y}{y^2}\right) \frac{2y}{y^2}$$

$$\Rightarrow y' = \frac{4y}{y^2} (xy' - 2y)$$

$$\Rightarrow y^3 = 4y (xy' - 2y)$$

[which is the required differential equation.]

Q

Find the DE of all straight line passing through the origin.

Solutions:

We know that the general equation of straight line is.

$$y - y_1 = m(x - x_1) \text{ ——— (1)}$$

Since (1) is passing through the origin i.e., (0,0) then,

$$y - 0 = m(x - 0)$$

Q Find the DE of all straight line not passing through the origin.

Solution:

We know that the general equation of straight line is  $y - y_1 = m(x - x_1)$  ——— ①

$$\Rightarrow y - y_1 = mx - mx_1 \text{ ——— ②}$$

Differentiating ① with respect to  $x$ , we get..

$$\frac{dy}{dx} = m \text{ ——— ③}$$

From ①, we get...

$$m = \frac{y - y_1}{x - x_1}$$

Putting this value of  $m$  in ③

$$\frac{dy}{dx} = \frac{y - y_1}{x - x_1}$$

which is the required Differential Equation.

# From a DE where  $y = a \cos x + b \sin x$   
where  $a, b$  are constant.

Solution:

Given that,  $y = a \cos x + b \sin x$  — ①

Differentiating ① w.r. to  $x$ , we get.

$$y' = -a \sin x + b \cos x$$

Again differentiating, we get.

$$y'' = -a \cos x - b \sin x$$

$$\Rightarrow y'' = -(a \cos x + b \sin x)$$

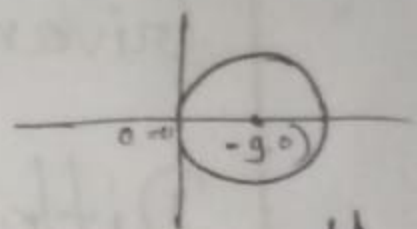
$$\Rightarrow y'' = -y \quad [\text{From ①}]$$

$$\Rightarrow y'' + y = 0$$

which is the required Differential Equation.

General Equation for Circle :  $x^2 + y^2 + 2gx + 2fy + c = 0$

# Find the DE of all circle passing through the origin & having their centers on the X-axis.



Solution:-

We know that the equation of circle that passes the origin & having their centers on the X-axis is.

$$x^2 + y^2 + 2gx = 0 \quad \text{--- (1)}$$

Differentiating (1) w.r.to  $x$ .

$$2x + 2yy' + 2g = 0$$

$$\Rightarrow x + yy' + g = 0$$

$$\Rightarrow g = -(x + yy') \quad \text{--- (2)}$$

putting this value into (1).

$$x^2 + y^2 + 2x(-x - yy') = 0$$

$$\Rightarrow x^2 + y^2 - 2x^2 - 2xyy' = 0$$

$$\Rightarrow y^2 - x^2 - 2xyy' = 0$$

which is the required DE.